Rebuttal response on theory.

Response to the theoretical concern. We contribute three results that together justify the design of filtered SFT with self-generation and refusal steering. **(1)** Self-generation is KL-optimal for reasoning preservation: the student's own safety-filtered distribution is the unique safe distribution minimizing forward KL to the student (Lemma 1). **(2)** Filtering is the least disruptive way to achieve perfect safety: it is the smallest χ^2 step from the source that places all mass on safe outputs (Proposition 1, specialized from Russo, 2026). **(3)** Refusal steering reduces generation cost without distorting the target: under a structural assumption (a label-only tilt), refusal steering preserves the accepted target *exactly* while providing up to an ω-fold speedup, with the largest gains on the hardest prompts (Corollary 1).

Setup and notation. Fix a prompt x and let $\mathcal{Y}$ denote the (finite) response space. Two models play distinct roles:

- the **student** $p_{\mathrm{ref}}(\cdot \mid x)$: the frozen model being safety-tuned;
- the **source** $\pi(\cdot \mid x)$: the model that generates candidate training responses (may equal p_{ref} for self-generation, or differ, e.g. a teacher).

A binary **safety filter** $\varphi(x, y) \in \{0, 1\}$ classifies each response as safe or unsafe. *Safety filtering* a model means conditioning it on safe outputs. For the source, define its acceptance rate and **safety-filtered distribution**:

$$\alpha_\pi(x) := \mathbb{P}_{y\sim\pi(\cdot|x)}[\varphi(x, y) = 1], \qquad \pi^+(y \mid x) := \frac{\pi(y \mid x)\,\varphi(x, y)}{\alpha_\pi(x)}.$$

For the student, define the same quantities:

$$\alpha_{\mathrm{ref}}(x) := \mathbb{P}_{y\sim p_{\mathrm{ref}}(\cdot|x)}[\varphi(x, y) = 1], \qquad p^+_{\mathrm{ref}}(y \mid x) := \frac{p_{\mathrm{ref}}(y \mid x)\,\varphi(x, y)}{\alpha_{\mathrm{ref}}(x)},$$

whenever $\alpha_{\mathrm{ref}}(x) > 0$.

Reasoning preservation. The KL divergence $D_{\mathrm{KL}}\big(\pi^+(\cdot \mid x) \,\|\, p_{\mathrm{ref}}(\cdot \mid x)\big)$ is a proxy for distributional discrepancy between the training target and the frozen student: for any bounded score $s(y) \in [0, 1]$, Pinsker's inequality gives

$$\Big|\mathbb{E}_{y\sim\pi^+}[s(y)] - \mathbb{E}_{y\sim p_{\mathrm{ref}}}[s(y)]\Big| \le \sqrt{\tfrac{1}{2} D_{\mathrm{KL}}\big(\pi^+(\cdot \mid x) \,\|\, p_{\mathrm{ref}}(\cdot \mid x)\big)},$$

so smaller KL means the safety-filtered source is less likely to shift the student's native reasoning.

1. Self-generation minimizes distributional discrepancy.

Lemma 1 (Student-relative safe projection)**.** *Assume $\alpha_{\mathrm{ref}}(x) > 0$. For any distribution $r(\cdot \mid x)$ supported on $\{y \in \mathcal{Y} : \varphi(x, y) = 1\}$,*

$$D_{\mathrm{KL}}\big(r(\cdot \mid x) \,\|\, p_{\mathrm{ref}}(\cdot \mid x)\big) = -\log \alpha_{\mathrm{ref}}(x) + D_{\mathrm{KL}}\big(r(\cdot \mid x) \,\|\, p^+_{\mathrm{ref}}(\cdot \mid x)\big).$$

Since the first term is independent of r and $D_{\mathrm{KL}}\big(r(\cdot \mid x) \,\|\, p^+_{\mathrm{ref}}(\cdot \mid x)\big) \ge 0$ with equality iff $r = p^+_{\mathrm{ref}}(\cdot \mid x)$, the unique safe distribution minimizing $D_{\mathrm{KL}}\big(r(\cdot \mid x) \,\|\, p_{\mathrm{ref}}(\cdot \mid x)\big)$ is $r^* = p^+_{\mathrm{ref}}(\cdot \mid x)$.

Applying Lemma 1 to $r = \pi^+(\cdot \mid x)$ yields:

$$D_{\mathrm{KL}}\big(\pi^+(\cdot \mid x) \,\|\, p_{\mathrm{ref}}(\cdot \mid x)\big) = \underbrace{-\log \alpha_{\mathrm{ref}}(x)}_{\text{unavoidable safe-filtering cost}} + \underbrace{D_{\mathrm{KL}}\big(\pi^+(\cdot \mid x) \,\|\, p^+_{\mathrm{ref}}(\cdot \mid x)\big)}_{\text{extra discrepancy from using source } \pi \text{ instead of } p_{\mathrm{ref}}}.$$

When the source is the student itself ($\pi = p_{\mathrm{ref}}$), $\pi^+ = p^+_{\mathrm{ref}}$ and the mismatch term vanishes: $D_{\mathrm{KL}}(\pi^+ \| p_{\mathrm{ref}}) = -\log \alpha_{\mathrm{ref}}$, the smallest KL attainable by *any* safe policy. A teacher source ($\pi = p_T$) incurs an additional cross-model gap $D_{\mathrm{KL}}(p^+_T \| p^+_{\mathrm{ref}}) \ge 0$.

2. Filtering is the least disruptive way to achieve perfect safety.

Proposition 1 (Source-centered policy improvement (one-step specialization of Russo, 2026))**.** *Assume $\alpha_\pi(x) > 0$. The accepted conditional $\pi^+(\cdot \mid x)$ is the unique optimizer of*

$$\max_{r\in\Delta(\mathcal{Y})} \mathbb{E}_{y\sim r}[\varphi(x, y)] \qquad \textit{subject to} \qquad \chi^2\big(r \,\|\, \pi(\cdot \mid x)\big) \le \frac{1 - \alpha_\pi(x)}{\alpha_\pi(x)}.$$

That is, filtering is the smallest χ^2-ball step from the source that achieves perfect safety reward.

For any source π, the safety-filtered distribution π^+ is the most conservative modification of π that places all mass on safe outputs. This holds regardless of whether the source is the student, a teacher, or a steered model.

3. Refusal steering reduces generation cost and eliminates gradient imbalance across harmful prompts.

Assumption 1 (Refusal tilt). *For a harmful prompt x_h, there exists $\omega(x_h) > 1$ such that the refusal instruction reweights safe outputs by $\omega(x_h)$ while leaving unsafe outputs unchanged:*

$$p_{\mathrm{ref}}(y \mid I_{\mathrm{refusal}}, x_h) \;\propto\; \begin{cases} \omega(x_h) \cdot p_{\mathrm{ref}}(y \mid x_h) & \text{if } \varphi(x_h, y) = 1, \\ p_{\mathrm{ref}}(y \mid x_h) & \text{if } \varphi(x_h, y) = 0. \end{cases}$$

That is, relative probabilities within the safe set and within the unsafe set are preserved; only the odds between the two groups change.

Corollary 1 (Refusal steering as cost-reducing preconditioning)**.** *Let $\pi_I(\cdot \mid x_h) := p_{\mathrm{ref}}(\cdot \mid I_{\mathrm{refusal}}, x_h)$ with acceptance rate $\alpha_I(x_h)$. Under Assumption 1,*

$$\pi_I^+(\cdot \mid x_h) = p_{\mathrm{ref}}^+(\cdot \mid x_h), \qquad \alpha_I(x_h) = \frac{\omega(x_h)\, \alpha_{\mathrm{ref}}(x_h)}{1 + \big(\omega(x_h) - 1\big)\, \alpha_{\mathrm{ref}}(x_h)} \geq \alpha_{\mathrm{ref}}(x_h),$$

with strict inequality when $\alpha_{\mathrm{ref}}(x_h) < 1$. Hence the accepted target is unchanged while the acceptance rate increases (strictly so whenever $0 < \alpha_{\mathrm{ref}}(x_h) < 1$). Collecting m accepted traces requires $\alpha_I(x_h)/\alpha_{\mathrm{ref}}(x_h) \geq 1$ times fewer expected generations with steering; for small $\alpha_{\mathrm{ref}}(x_h)$ this speedup approaches $\omega(x_h)$.

Since $\pi_I^+ = p_{\mathrm{ref}}^+$ exactly, refusal steering achieves the same KL minimum as benign self-generation ($\pi = p_{\mathrm{ref}}$): $D_{\mathrm{KL}}(\pi_I^+ \| p_{\mathrm{ref}}) = -\log \alpha_{\mathrm{ref}}(x_h)$. Moreover, prompts with lower acceptance rates receive a *larger* multiplicative boost: for any $0 < a \leq b \leq 1$ and common $\omega > 1$, $f_\omega(a)/f_\omega(b) > a/b$ where $f_\omega(a) := \omega a / \big(1 + (\omega - 1)a\big)$. Thus steering can reduce the *multiplicative* acceptance-rate imbalance across harmful prompts, especially when ω is large enough to raise all rates above a common threshold.

Appendix: proof details and relation to Russo (2026)

For a fixed prompt x, let $\mathcal{Y}$ be a finite response space. Let $\pi(\cdot \mid x)$ denote the source policy used to generate candidate responses and let $\varphi(x, y) \in \{0, 1\}$ be the safety filter. Define

$$\alpha_\pi(x) := \mathbb{P}_{y\sim\pi(\cdot|x)}[\varphi(x, y) = 1], \qquad \pi^+(y \mid x) := \frac{\pi(y \mid x)\varphi(x, y)}{\alpha_\pi(x)},$$

and assume $\alpha_\pi(x) > 0$. Also define

$$\alpha_{\mathrm{ref}}(x) := \mathbb{P}_{y\sim p_{\mathrm{ref}}(\cdot|x)}[\varphi(x, y) = 1], \qquad p^+_{\mathrm{ref}}(y \mid x) := \frac{p_{\mathrm{ref}}(y \mid x)\varphi(x, y)}{\alpha_{\mathrm{ref}}(x)},$$

and assume $\alpha_{\mathrm{ref}}(x) > 0$.

Lemma 1 (Student-relative safe projection)**.** *For any distribution $r(\cdot \mid x)$ supported on $\{y \in \mathcal{Y} : \varphi(x, y) = 1\}$,*

$$D_{\mathrm{KL}}\big(r(\cdot \mid x) \,\|\, p_{\mathrm{ref}}(\cdot \mid x)\big) = -\log \alpha_{\mathrm{ref}}(x) + D_{\mathrm{KL}}\big(r(\cdot \mid x) \,\|\, p^+_{\mathrm{ref}}(\cdot \mid x)\big).$$

Since the first term is independent of r and $D_{\mathrm{KL}}\big(r(\cdot \mid x) \,\|\, p^+_{\mathrm{ref}}(\cdot \mid x)\big) \geq 0$ with equality iff $r = p^+_{\mathrm{ref}}(\cdot \mid x)$, the unique safe distribution minimizing $D_{\mathrm{KL}}\big(r(\cdot \mid x) \,\|\, p_{\mathrm{ref}}(\cdot \mid x)\big)$ is $r^* = p^+_{\mathrm{ref}}(\cdot \mid x)$.

Proof. Because $r(\cdot \mid x)$ is supported on the accepted set, we have

$$p_{\mathrm{ref}}(y \mid x) = \alpha_{\mathrm{ref}}(x)\, p^+_{\mathrm{ref}}(y \mid x) \qquad \text{for all } y \text{ in the support of } r(\cdot \mid x).$$

Therefore,

$$\begin{aligned}
D_{\mathrm{KL}}\big(r(\cdot \mid x) \,\|\, p_{\mathrm{ref}}(\cdot \mid x)\big) &= \sum_{y\in\mathcal{Y}} r(y \mid x) \log \frac{r(y \mid x)}{p_{\mathrm{ref}}(y \mid x)} \\
&= \sum_{y\in\mathcal{Y}} r(y \mid x) \log \frac{r(y \mid x)}{\alpha_{\mathrm{ref}}(x) p^+_{\mathrm{ref}}(y \mid x)} \\
&= -\log \alpha_{\mathrm{ref}}(x) + \sum_{y\in\mathcal{Y}} r(y \mid x) \log \frac{r(y \mid x)}{p^+_{\mathrm{ref}}(y \mid x)} \\
&= -\log \alpha_{\mathrm{ref}}(x) + D_{\mathrm{KL}}\big(r(\cdot \mid x) \,\|\, p^+_{\mathrm{ref}}(\cdot \mid x)\big).
\end{aligned}$$

The minimization claim follows because KL divergence is nonnegative and equals zero if and only if the two distributions are equal. □

Proposition 1 (Source-centered improvement and equivalent KL identities)**.** *Fix a prompt x and treat each complete response $y \in \mathcal{Y}$ as a one-step action with binary reward $\varphi(x, y)$. Then:*

(a) The accepted conditional $\pi^+(\cdot \mid x)$ is the unique optimizer of

$$\max_{r\in\Delta(\mathcal{Y})} \mathbb{E}_{y\sim r}[\varphi(x, y)] \qquad \textit{subject to} \qquad \chi^2\big(r \,\|\, \pi(\cdot \mid x)\big) \leq \frac{1 - \alpha_\pi(x)}{\alpha_\pi(x)},$$

where

$$\chi^2\big(r \,\|\, \pi(\cdot \mid x)\big) := \sum_{y\in\mathcal{Y}} \pi(y \mid x) \left(\frac{r(y)}{\pi(y \mid x)} - 1\right)^2,$$

with the convention that the divergence is $+\infty$ if r is not absolutely continuous with respect to $\pi(\cdot \mid x)$. This is the one-step finite-response specialization of Russo (2026, Proposition 4.1); the same Cauchy–Schwarz step appears in Russo's Appendix A.3.

(b) The KL shift from the frozen student satisfies

$$D_{\mathrm{KL}}\big(\pi^+(\cdot \mid x) \,\|\, p_{\mathrm{ref}}(\cdot \mid x)\big) = -\log \alpha_{\mathrm{ref}}(x) + D_{\mathrm{KL}}\big(\pi^+(\cdot \mid x) \,\|\, p^+_{\mathrm{ref}}(\cdot \mid x)\big).$$

Equivalently,

$$D_{\mathrm{KL}}\big(\pi^+(\cdot \mid x) \,\|\, p_{\mathrm{ref}}(\cdot \mid x)\big) = -\log \alpha_\pi(x) + \mathbb{E}_{y\sim\pi^+(\cdot|x)}\left[\log \frac{\pi(y \mid x)}{p_{\mathrm{ref}}(y \mid x)}\right],$$

with

$$\mathbb{E}_{y\sim\pi^+(\cdot|x)}\left[\log \frac{\pi(y \mid x)}{p_{\mathrm{ref}}(y \mid x)}\right] = \log \frac{\alpha_\pi(x)}{\alpha_{\mathrm{ref}}(x)} + D_{\mathrm{KL}}\big(\pi^+(\cdot \mid x) \,\|\, p^+_{\mathrm{ref}}(\cdot \mid x)\big).$$

Proof. **Part (a).** This is the one-step finite-response specialization of Russo (2026, Proposition 4.1), and the same proof pattern appears in Russo's Appendix A.3. For completeness, let

$$S_x := \{y \in \mathcal{Y} : \varphi(x, y) = 1\}.$$

Any optimizer must achieve reward 1 and hence be supported on S_x. For any such r,

$$\chi^2\big(r \,\|\, \pi(\cdot \mid x)\big) = \sum_{y \in S_x} \frac{r(y)^2}{\pi(y \mid x)} - 1,$$

and Cauchy–Schwarz gives

$$\left(\sum_{y \in S_x} r(y)\right)^2 \le \left(\sum_{y \in S_x} \pi(y \mid x)\right)\left(\sum_{y \in S_x} \frac{r(y)^2}{\pi(y \mid x)}\right),$$

so

$$\sum_{y \in S_x} \frac{r(y)^2}{\pi(y \mid x)} \ge \frac{1}{\alpha_\pi(x)} \qquad \Longrightarrow \qquad \chi^2\big(r \,\|\, \pi(\cdot \mid x)\big) \ge \frac{1 - \alpha_\pi(x)}{\alpha_\pi(x)}.$$

Equality holds if and only if $r(y) \propto \pi(y \mid x)$ on S_x, which after normalization gives $r = \pi^+$. Since $\chi^2\big(\pi^+(\cdot \mid x) \,\|\, \pi(\cdot \mid x)\big) = (1 - \alpha_\pi(x))/\alpha_\pi(x)$, π^+ is the unique optimal feasible distribution.

Part (b). The first identity is Lemma 1 with $r = \pi^+$. For the equivalent source expansion, note that on the support of $\pi^+(\cdot \mid x)$,

$$\pi(y \mid x) = \alpha_\pi(x)\, \pi^+(y \mid x), \qquad p_{\mathrm{ref}}(y \mid x) = \alpha_{\mathrm{ref}}(x)\, p^+_{\mathrm{ref}}(y \mid x).$$

Therefore,

$$\log \frac{\pi(y \mid x)}{p_{\mathrm{ref}}(y \mid x)} = \log \frac{\alpha_\pi(x)}{\alpha_{\mathrm{ref}}(x)} + \log \frac{\pi^+(y \mid x)}{p^+_{\mathrm{ref}}(y \mid x)}.$$

Averaging under $\pi^+(\cdot \mid x)$ yields

$$\mathbb{E}_{y \sim \pi^+(\cdot \mid x)}\left[\log \frac{\pi(y \mid x)}{p_{\mathrm{ref}}(y \mid x)}\right] = \log \frac{\alpha_\pi(x)}{\alpha_{\mathrm{ref}}(x)} + D_{\mathrm{KL}}\big(\pi^+(\cdot \mid x) \,\|\, p^+_{\mathrm{ref}}(\cdot \mid x)\big),$$

which rearranges to the second displayed identity. □

Note. The equivalent source expansion is algebraically correct, but its second term is not sign-definite in general; the sign-definite decomposition is the projection form through $p^+_{\mathrm{ref}}(\cdot \mid x)$.

Assumption 1 (Refusal tilt). For a harmful prompt x_h, suppose there exists $\omega(x_h) > 1$ such that the refusal instruction reweights safe outputs by $\omega(x_h)$ while leaving unsafe outputs unchanged:

$$p_{\mathrm{ref}}(y \mid I_{\mathrm{refusal}}, x_h) \;\propto\; \begin{cases} \omega(x_h) \cdot p_{\mathrm{ref}}(y \mid x_h) & \text{if } \varphi(x_h, y) = 1, \\ p_{\mathrm{ref}}(y \mid x_h) & \text{if } \varphi(x_h, y) = 0. \end{cases}$$

That is, relative probabilities within the safe set and within the unsafe set are preserved; only the odds between the two groups change. The normalizing constant is $Z(x_h) = 1 + \big(\omega(x_h) - 1\big)\, \alpha_{\mathrm{ref}}(x_h)$.

Corollary 1 (Refusal steering as cost-reducing preconditioning)**.** *Let $\pi_I(\cdot \mid x_h) := p_{\mathrm{ref}}(\cdot \mid I_{\mathrm{refusal}}, x_h)$ and $\alpha_I(x_h) := \mathbb{P}_{y \sim \pi_I}[\varphi(x_h, y) = 1]$. Under Assumption 1:*

(a) $\pi^+_I(\cdot \mid x_h) = p^+_{\mathrm{ref}}(\cdot \mid x_h)$ *and* $\alpha_I(x_h) = \dfrac{\omega(x_h)\, \alpha_{\mathrm{ref}}(x_h)}{1 + \big(\omega(x_h) - 1\big)\, \alpha_{\mathrm{ref}}(x_h)}$.

(b) Let T^{base}_m and T^{steer}_m denote the numbers of raw generations needed to collect m accepted traces under $p_{\mathrm{ref}}(\cdot \mid x_h)$ and $\pi_I(\cdot \mid x_h)$ respectively. Then

$$\mathbb{E}[T^{\mathrm{base}}_m] = \frac{m}{\alpha_{\mathrm{ref}}(x_h)}, \qquad \mathrm{Var}(T^{\mathrm{base}}_m) = \frac{m\big(1 - \alpha_{\mathrm{ref}}(x_h)\big)}{\alpha_{\mathrm{ref}}(x_h)^2},$$

$$\mathbb{E}[T^{\mathrm{steer}}_m] = \frac{m}{\alpha_I(x_h)}, \qquad \mathrm{Var}(T^{\mathrm{steer}}_m) = \frac{m\big(1 - \alpha_I(x_h)\big)}{\alpha_I(x_h)^2}.$$

Moreover, for the mean ratio,

$$\frac{\mathbb{E}[T_m^{\text{steer}}]}{\mathbb{E}[T_m^{\text{base}}]} = \alpha_{\text{ref}}(x_h) + \frac{1 - \alpha_{\text{ref}}(x_h)}{\omega(x_h)},$$

and when $0 < \alpha_{\text{ref}}(x_h) < 1$,

$$\frac{\text{Var}(T_m^{\text{steer}})}{\text{Var}(T_m^{\text{base}})} \leq \frac{1}{\omega(x_h)}.$$

Thus refusal steering acts as a cost-reducing proposal distribution for filtered SFT: it leaves the accepted target p_{ref}^+ *unchanged but lowers the sampling cost needed to collect accepted traces.*

Proof. **Part (a).** Write $Z = 1 + (\omega - 1)\alpha_{\text{ref}}$ (suppressing x_h). For y with $\varphi(x_h, y) = 1$, the assumption gives $\pi_I(y \mid x_h) = \omega\, p_{\text{ref}}(y \mid x_h)/Z$. Summing over the safe set yields $\alpha_I = \omega\, \alpha_{\text{ref}}/Z$. The accepted conditional is

$$\pi_I^+(y \mid x_h) = \frac{\pi_I(y \mid x_h)\, \varphi(x_h, y)}{\alpha_I} = \frac{\omega\, p_{\text{ref}}(y \mid x_h)/Z}{\omega\, \alpha_{\text{ref}}/Z} = \frac{p_{\text{ref}}(y \mid x_h)}{\alpha_{\text{ref}}} = p_{\text{ref}}^+(y \mid x_h).$$

Part (b). Collecting m accepted traces by rejection sampling requires $T_m = \sum_{i=1}^m G_i$ where each $G_i \sim \text{Geometric}(\alpha)$, giving $\mathbb{E}[T_m] = m/\alpha$ and $\text{Var}(T_m) = m(1-\alpha)/\alpha^2$.
Mean ratio. $\mathbb{E}[T_m^{\text{steer}}]/\mathbb{E}[T_m^{\text{base}}] = \alpha_{\text{ref}}/\alpha_I = Z/\omega = 1/\omega + (1 - 1/\omega)\alpha_{\text{ref}} = \alpha_{\text{ref}} + (1 - \alpha_{\text{ref}})/\omega$.
Variance ratio. From part (a), $1 - \alpha_I = (1 - \alpha_{\text{ref}})/Z$, so

$$\frac{\text{Var}(T_m^{\text{steer}})}{\text{Var}(T_m^{\text{base}})} = \frac{(1 - \alpha_I)\, \alpha_{\text{ref}}^2}{(1 - \alpha_{\text{ref}})\, \alpha_I^2} = \frac{\alpha_{\text{ref}}^2}{Z\, \alpha_I^2} = \frac{Z}{\omega^2} = \frac{1 + (\omega - 1)\alpha_{\text{ref}}}{\omega^2}.$$

Since $\alpha_{\text{ref}} \leq 1$, the numerator is at most ω, yielding the bound $1/\omega$.
Concrete example. If $\alpha_{\text{ref}} = 0.05$, then $\text{Var}(T_m^{\text{base}}) = 380\, m$. If steering raises acceptance above 0.5 (i.e. $\omega \geq 19$), then $\text{Var}(T_m^{\text{steer}}) \leq 2\, m$—a roughly 190-fold reduction. □

References

[1] D. Russo. Success Conditioning as Policy Improvement: The Optimization Problem Solved by Imitating Success. *arXiv preprint arXiv:2601.18175*, 2026.